llmXive Automated Review of Video Generation Models are General-Purpose Vision Learners

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several strong claims regarding state-of-the-art (SOTA) performance and data efficiency that require closer alignment with the provided tables.

First, the abstract and introduction assert that the model achieves comparable performance to leading models like D4RT and VGGT-Ω with "7× to 500× less training data." Table 2 provides the data counts: Ours (14B) uses 1.23M frames, while VGGT-Ω uses ~600M frames (a ~487× difference) and D4RT uses ~86M frames (a ~70× difference). The upper bound of "500×" is not strictly supported by the numbers in Table 2 (max ~487×). This should be corrected to "up to ~490×" or the specific comparison yielding 500× must be identified.

Second, the claim of outperforming "DepthAnything V3, SAM3, D4RT..." across tasks is partially contradicted by Table 1. While the model outperforms DepthAnything 3 on Sintel and KITTI, it underperforms on ETH3D (0.044 vs 0.028 AbsRel). The blanket statement "outperforms" is inaccurate; it should be qualified to reflect mixed results or specific benchmarks. Additionally, the claim of outperforming SAM3 on depth/normal tasks is unsupported as Table 1 does not list SAM3 results for these specific metrics.

Third, the bibliography cites "VGGT-Ω" (2026) and "SAM3" (2025) as established baselines. As these are future-dated relative to the current review context, the validity of the "SOTA" claim depends on the existence and reproducibility of these specific future models. If these are hypothetical or concurrent works not yet public, the comparison should be framed as such to avoid misleading readers about the current state of the art.

Finally, the claim of "comparable performance" with VGGT-Ω in the context of data efficiency is weakened by the performance gap in Table 2 (0.093 vs 0.067 Average AbsRel). A 30% higher error rate is significant; the text should clarify if "comparable" refers to a specific metric or if the data efficiency trade-off is explicitly acknowledged as a performance drop.

Action items.

- [**writing**] The paper makes several strong claims regarding state-of-the-art (SOTA) performance and data efficiency that require closer alignment with the provided tables. First, the abstract and introduction assert that the model achieves comparable performance to leading models like D4RT and VGGT-Ω with "7× to 500× less training data." Table 2 provides the data counts: Ours (14B) uses 1.23M frames, while VGGT-Ω uses ~600M frames (a ~487× difference) and D4RT uses ~86M frames (a ~70× difference). The upper

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 effectively visualizes the methodology and paradigm shift, but the caption contains a spelling error ('Methdology') and grammatical omissions that leave sentences without clear subjects.

Figure 2

Figure 2 presents a radar chart and a scatter plot to demonstrate generalist capability and data efficiency. However, the caption is truncated, and the right panel's y-axis break is visually confusing, potentially misrepresenting the performance gap between models. The radar chart also lacks a formal legend for the various competitor models.

Figure 3

Figure 3 is a teaser-style visualization that is rendered clearly but is scientifically unusable due to a complete lack of a descriptive caption and internal labels. The viewer cannot identify the specific tasks or data modalities shown without guessing.

Figure 4

The figure provides a clear visual overview of the architecture, but the caption contains significant grammatical errors and appears to include a sentence fragment from another figure's description.

Figure 5

Figure 5 visually depicts a transformation from separate rotation and translation maps to a combined 'Rothko' map, but lacks any quantitative or qualitative legend to explain the color encoding or channel composition described in the caption.

Figure 6

The figure effectively demonstrates depth and normal estimation capabilities with clear visual separation between RGB, depth, and normal outputs. However, the caption contains a missing model name, and the surface normal visualization lacks a legend to define the color mapping.

Figure 7

The figure provides clear qualitative examples of segmentation masks for various prompts, but the static presentation fails to visually demonstrate the 'spatial relationships' and 'motion' reasoning claimed in the caption.

Figure 8

The figure effectively demonstrates the impact of layer transfer on training loss, but the legend contains inconsistent grammar ('layer' vs 'layers') and a minor typo in the final entry.

Figure 9

The figure visually demonstrates the model's ability to handle multiple people, but the caption is grammatically fragmented and includes a panel label '(a)' that does not correspond to the single image grid shown.

Action items.

- [**writing**] Figure 1: The caption contains a typo 'Methdology' and omits the model name (e.g., 'Our model') in the first sentence, making the text grammatically incomplete.
- [**writing**] Figure 1: The caption text 'shows strong performance' lacks a subject, failing to explicitly link the 'Emerging Behaviors' section to the model.
- [**writing**] Figure 2: The caption text is truncated mid-sentence at the end ('...outperforms the largest available variants alter [results_combined.png]'), leaving the description of the Data Efficiency plot incomplete.
- [**science**] Figure 2: The right panel's y-axis contains a visual break (indicated by jagged lines) between 0.16 and 0.26, but the data points in that range (e.g., 'Ours (WAN 1.3B, 0.9M frames)' at ~0.125) are plotted below the break, creating a misleading visual gap and distorting the perceived performance difference between the top and bottom clusters.
- [**writing**] Figure 2: The left radar chart lacks a clear legend defining the specific colors/shapes for the competitor models (e.g., D4RT, Sapiens-2B, DepthAnythingV3), relying solely on direct labels which can be cluttered and hard to distinguish.
- [**fatal**] Figure 3: The caption is explicitly '(no caption)', providing no context for the visual content. Without a description, it is impossible to determine what the different columns (e.g., depth maps, segmentation masks, skeleton overlays) represent or what the figure is intended to demonstrate.
- [**science**] Figure 3: The figure lacks any axis labels, units, or legends to explain the color mappings in the intermediate columns (e.g., depth, surface normals) or the specific tasks being visualized in the rightmost columns.
- [**writing**] Figure 4: The caption contains a grammatical error and missing subject: 'Architecture overview of , a simple yet powerful architecture...' (missing model name).
- [**writing**] Figure 4: The caption contains a sentence fragment: 'During multi-task post-training, the model is adapted to feed-forward model fine-tuned on predominantly synthetic data to handle diverse perception tasks.' This appears to be a copy-paste error from Figure 1's caption.
- [**science**] Figure 4: The diagram shows 'Learnable Tokens' entering the Pretrained DiT, but the caption states sparse tasks are realized by adding them as 'additional inputs to the diffusion transformer (DiT)'. The visual flow is clear, but the caption phrasing is slightly ambiguous regarding whether they are added to the input or the transformer layers.
- [**science**] Figure 5: The caption claims the 'Rothko' Raymap assembles rotation and translation components into a single three-channel map, but the image shows a green rectangle

(translation) embedded in a gradient (rotation) without any indication of how these are combined into RGB channels or what the color mapping represents.

- [**writing**] Figure 5: No colorbar, legend, or axis labels are present to explain the meaning of the colors or spatial dimensions in either the 'Rotation Raymap' or 'Rothko Raymap' visualizations.
- [**writing**] Figure 6 caption contains a grammatical error ('Demonstration of 's depth...') where the model name is missing before the possessive.
- [**science**] Figure 6: The rightmost column displays surface normals, but the image is rendered with a distinct color map (purple/green) compared to the middle column (orange/blue). Without a legend or explicit label, it is difficult to verify if the color mapping corresponds to the correct normal vector components.
- [**writing**] Figure 7: The caption claims the model recognizes 'spatial relationships' and 'motion', but the figure only displays static input frames and predicted masks without visualizing the temporal or spatial reasoning process (e.g., no arrows, trajectory lines, or multi-frame sequences).
- [**writing**] Figure 7: The 'Emerging Behavior' section includes a prompt stating 'rocket' is not in training data, but the figure lacks a visual comparison or baseline to substantiate the claim of generalization to unseen objects.
- [**writing**] Figure 8: The legend uses inconsistent singular/plural phrasing ('0 pretrained layer' vs '8 pretrained layer'); standardize to 'layers' for all entries.
- [**writing**] Figure 8: The legend entry '40 pretrained layer (fully pretrained model)' contains a typo ('fully pretrained' should be 'fully pretrained').
- [**writing**] Figure 9: The caption '(a) Generalize to multiple instances' implies a multi-panel figure, but only a single image grid is shown; remove the '(a)' or add the missing panel.
- [**writing**] Figure 9: The caption is grammatically incomplete and lacks a subject (e.g., 'Our model generalizes...'); it currently reads as a fragment.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally accessible to a competent reader from an adjacent field, successfully bridging generative modeling and perception. However, several instances of subfield-specific shorthand and undefined acronyms create barriers.

In Section 3.1, the definition of velocity v in the Rectified Flow context is assumed rather than explicitly stated, leading to potential confusion regarding the sign convention when negating the output. In Section 3.2, terms like "raymap" and "RoPE" are used without expansion; while standard in their respective niches (3D reconstruction and LLMs), they are not universal to a general vision audience. Section 3.3 uses "HDRI" without expansion. Section 4.1 introduces "gradient dropping," which is non-standard and likely a misnomer for batch rejection, potentially confusing readers familiar with standard optimization. Finally, Section 4.3 uses "J&F" without defining the metric components.

Addressing these with brief parenthetical expansions or clarifications will ensure the paper is self-contained for the target "adjacent-field PhD" audience without diluting technical precision.

Action items.

- [**writing**] Section 3.1: Define the specific Rectified Flow velocity formulation ($v = psilon - x_0$) explicitly to clarify the sign convention used when negating the output, as this varies across literature.
- [**writing**] Section 3.2: Expand 'raymap' to 'pixel-space raymap (encoding 3D ray origins and directions)' to ensure readers from non-3D backgrounds understand the data representation.
- [**writing**] Section 3.2: Expand 'RoPE' to 'Rotary Positional Embeddings (RoPE)' at first use, as it is LLM-specific terminology not universal to all vision subfields.
- [**writing**] Section 3.3: Expand 'HDRI' to 'High Dynamic Range Image (HDRI)' to ensure clarity for readers unfamiliar with 3D graphics terminology.
- [**writing**] Section 4.1: Clarify 'gradient dropping' as 'batch rejection based on gradient norm' or confirm if this is a novel term, as it conflicts with standard optimization terminology.
- [**writing**] Section 4.3: Expand 'J&F' to 'Jaccard index and F-score (mean of IoU and contour accuracy)' at first mention to aid readers outside video segmentation.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent argument that video generation pre-training serves as a superior foundation for general-purpose vision tasks. The methodology logically follows from the premise that generative models encode rich spatiotemporal priors. However, there are specific inconsistencies between the textual claims and the reported data that undermine the internal consistency of the results section.

First, in Section 4.3 (Ablation Studies), the authors state: "Most notably, we empirically observe that joint training severely degrades 3D human keypoint estimation." This conclusion is drawn to explain why the generalist model might struggle with sparse tasks. However, Table 1 presents data that contradicts this claim. The "Ours - Generalist - L" model achieves an MPJPE of 71.8 on the EMDB dataset. This is identical to the "Ours - Specialist - L" (71.8) and significantly better than the "Ours - Specialist - S" (72.6) and the specialist baseline "Genmo" (73.0). If the generalist matches the best specialist and outperforms the smaller specialist, the claim of "severe degradation" is not supported by the provided numbers. The text suggests a failure mode that the table does not show. This requires either a correction of the text to reflect that performance is maintained (not degraded) or a re-evaluation of the data if the table values are incorrect.

Second, the claim of "7× to 500× less training data" appears in the Abstract, Introduction, and Section 4.3. Table 2 provides the specific numbers for the comparison: VGGT-Ω uses ~3M videos (~600M frames), while the proposed method uses 8.08K videos (1.23M frames). The ratio of frames is approximately 487x, and the ratio of videos is approximately 371x. While these fall within the 7x-500x range, the lower bound of "7x" is not accounted for in this specific comparison (the minimum ratio here is ~371x). The "7x" figure likely refers to a comparison with a different model (perhaps D4RT, which uses ~1M videos) or a specific subset of data not explicitly detailed in the table's summary row. Without clarifying which specific comparison yields the 7x figure, the range "7x to 500x" appears to be an aggregation of disparate comparisons that obscures the actual data efficiency relative to the specific baselines listed in the table. The text should specify the baseline for the lower bound or adjust the range to match the data presented in Table 2.

These issues do not invalidate the core hypothesis but represent a break in the logical chain between the evidence presented (tables) and the conclusions drawn (text).

Action items.

- [**science**] Section 4.3 claims joint training 'severely degrades' 3D keypoint estimation, but Table 1 shows Generalist-L (71.8 MPJPE) matches Specialist-L (71.8) and beats Specialist-S (72.6). The text contradicts the table data; clarify the baseline or correct the claim.
- [**writing**] The '7x to 500x' data efficiency claim in Abstract/Intro lacks a clear source for the 7x lower bound in Table 2, where the video ratio is ~371x and frame ratio ~487x vs VGGT-Ω. Specify which comparison yields the 7x figure or adjust the range to match the presented data.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes ambitious claims about achieving a "general-purpose" vision model and a "paradigm shift" toward "general visual intelligence," but the empirical evidence provided is scoped narrowly to specific tasks and domains.

First, the Abstract and Introduction assert that the method "solves" the problem of generalist vision. However, Table 1 reveals that the "Generalist" variant degrades performance on 3D human keypoint estimation compared to the "Specialist" variant, a fact admitted in the Ablation

Studies. The claim of “solving” the problem is too strong given these specific failures and the narrow scope of evaluation.

Second, the Abstract and Figure 2 caption claim the model “universally” outperforms specialists and generalizes to “any” object category. The data contradicts the “universal” claim: the model was trained almost exclusively on synthetic human data, and quantitative benchmarks are heavily human-centric. While qualitative examples show generalization to animals and robots, these lack quantitative metrics on standard benchmarks for those categories. The claim of generalization to “any” category overreaches the evidence.

Finally, the Conclusion states the work provides “strong evidence for the existence of a universal ‘world model’.” The experiments validate performance on geometric and segmentation tasks but do not test the model’s ability to reason about physics or causality in a general sense. The term “world model” implies a broader understanding than what is demonstrated.

To address these issues, the authors should:

1. Replace absolute terms like “solves,” “universally,” and “any” with precise qualifiers.
2. Explicitly acknowledge the limitations of the evaluation scope (human-centric, specific tasks) in the Abstract and Conclusion.
3. Clarify that the “generalization to animals/robots” is currently qualitative and requires further quantitative validation.

Action items.

- [**writing**] Abstract claims the method ‘solves’ the quest for general-purpose vision, but Table 1 shows performance degradation on 3D keypoint estimation in the generalist setting. Replace ‘solves’ with ‘advances’ and qualify the scope to tested tasks.
- [**writing**] Abstract and Fig 2 claim ‘universal’ performance and generalization to ‘any’ category, yet evaluation is limited to human-centric synthetic data and specific benchmarks. Qualitative examples of animals/robots lack quantitative backing. Narrow claims to ‘demonstrates zero-shot transfer in examples’.
- [**writing**] Conclusion claims ‘strong evidence for a universal world model,’ but experiments only test geometry and pose, not physics or causality. Rephrase to ‘evidence that video backbones encode priors useful for specific perception tasks’.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

This paper presents a method for repurposing pre-trained video generation models for general-purpose vision perception tasks. The methodology relies on fine-tuning a large-scale diffusion backbone (WAN 2.1) using a dataset of 7,500 synthetic videos generated from 3D assets (RenderPeople) and motion capture data (CMU).

From a safety and ethics perspective, the work is low-risk. The primary data source is synthetic, generated from licensed 3D assets and public motion capture data, which effectively mitigates risks associated with PII, consent, and copyright infringement regarding real human subjects. The paper explicitly states in Section 4.3 that the model is trained on synthetic data and generalizes to real-world footage, but the training data itself does not contain identifiable individuals.

The paper does not describe any dual-use capabilities that lower the barrier to harm (e.g., generating deepfakes, surveillance tools, or cyber-attacks). Instead, it focuses on perception tasks (depth estimation, segmentation, pose estimation) which are generally beneficial. While the underlying video generation model (WAN 2.1) could theoretically be used for synthesis, this paper's contribution is the *repurposing* of that model for analysis/perception, not the generation of deceptive content. No operational details for exploits or vulnerabilities are disclosed.

There are no missing disclosures regarding human subjects, as the data is synthetic. There are no indications of bias against specific demographic groups in the training data (which is synthetic and human-centric but not tied to real identities). The paper does not require a specific ethics statement beyond the standard description of data provenance, which is provided.

Consequently, there are no foreseeable, non-trivial risks of harm that the paper fails to acknowledge or mitigate. The work is safe to publish as is.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling hypothesis that video generation models serve as superior pre-training backbones for general-purpose vision tasks. However, the evidentiary support for the central claims of "state-of-the-art" performance and "exceptional data efficiency" is currently weakened by a lack of statistical rigor and potential confounds in the experimental design.

First, the quantitative results in Table 1 and Table 2 are presented as single-point estimates without any indication of variance (standard deviation, confidence intervals) or the number of random seeds used. In deep learning benchmarks, especially with large models, performance can fluctuate by 1-2% across seeds. A claimed improvement of 0.01 in AbsRel or a few points in J&F could easily be within the noise floor. Without reporting mean $\pm$ std over at least 3-5 seeds, the reader cannot distinguish a genuine architectural advantage from a lucky initialization or a specific random seed.

Second, the "data efficiency" claim (achieving comparable performance with 7x to 500x less training data) is potentially confounded by the training data composition. The proposed method is trained on a specific synthetic dataset (~1.2M frames), while the strong baselines

(D4RT, VGGT-Ω) are trained on massive, diverse real-world datasets (~86M-600M frames). The performance gap might stem from the *quality* or *diversity* of the synthetic data or the specific pre-training backbone (WAN 2.1) rather than the "generative pre-training" paradigm itself. To isolate the contribution of the generative prior, the authors should include a control experiment: fine-tune the same WAN 2.1 backbone on the *same* massive dataset used by the baselines. If the performance does not scale proportionally with the data volume in this control, the "data efficiency" claim is stronger; if it does, the claim may be overstated.

Finally, the "emergent behaviors" regarding generalization to out-of-distribution (OOD) categories (e.g., animals, robots) are supported only by qualitative visual examples in Figure 4. While visually impressive, this is insufficient to rule out that the model is simply memorizing specific textures or lighting conditions present in the synthetic training data that coincidentally match the test set. To substantiate this claim, the authors must provide quantitative metrics (e.g., MPJPE, IoU) on a standard, held-out OOD benchmark (such as EMDB for non-human subjects) to demonstrate that the generalization is robust and not an artifact of the specific test images chosen.

Addressing these points—specifically by adding variance reporting, a data-matched control for the efficiency claim, and quantitative OOD evaluation—would significantly strengthen the scientific validity of the paper's central contributions.

Action items.

- [**writing**] The paper presents a compelling hypothesis that video generation models serve as superior pre-training backbones for general-purpose vision tasks. However, the evidentiary support for the central claims of "state-of-the-art" performance and "exceptional data efficiency" is currently weakened by a lack of statistical rigor and potential confounds in the experimental design. First, the quantitative results in Table 1 and Table 2 are presented as single-point estimates without any indication of var

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical treatment of the quantitative results in this paper is insufficient to support the strong claims of "state-of-the-art" performance and "significant" improvements. While the experimental design is sound, the analysis lacks necessary rigor regarding uncertainty and inference.

First, **uncertainty reporting is missing**. Tables 1 and 2 present performance metrics (e.g., AbsRel, MPJPE) as single point estimates. In deep learning, performance varies significantly across random seeds. Reporting a single number implies unjustified precision. The authors must report the mean and standard deviation across at least 3 independent training runs for all metrics. Without this, it is impossible to determine if observed differences are robust or artifacts of initialization.

Second, **statistical significance is asserted without evidence**. The abstract and Section 5.1 use terms like "significantly better" based solely on numerical ordering. No formal hypothesis tests (e.g., paired t-tests) are described. The paper must either conduct appropriate tests and report p-values or soften the language to "numerically higher/lower" to avoid implying statistical rigor where none exists.

Third, **multiple comparisons are unaddressed**. The paper evaluates the model across numerous tasks and benchmarks, effectively performing dozens of pairwise comparisons. Highlighting the "best" results without correcting for multiplicity (e.g., Bonferroni or FDR) inflates the false-positive rate. The authors should apply a correction method or explicitly acknowledge the multiplicity issue when interpreting the results.

These issues are critical for establishing the credibility of the quantitative claims. The fix requires re-running experiments with multiple seeds and updating the text to reflect proper statistical inference.

Action items.

- [**science**] Tables 1 and 2 report single point estimates without uncertainty (SD/SE/CI). Deep learning results vary by seed; report mean ± SD over 3 seeds for all metrics to assess stability.
- [**writing**] Claims of 'significant' improvement (Abstract, Sec 5.1) rely on point estimates without hypothesis tests. Run paired t-tests/bootstrap, report p-values, or rephrase to 'numerically better'.
- [**science**] Table 1 compares 4 baselines across 6 benchmarks (24 tests) with no multiple-comparison correction. Apply Holm/BH correction or explicitly acknowledge the high false-positive risk in the text.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The paper is generally well-structured and the core argument flows logically from the problem statement to the proposed solution. However, several sentences suffer from redundancy, awkward phrasing, or minor grammatical errors that force the reader to pause and re-parse.

In the Introduction, the authors repeat the concept of "unified" and "model" in close proximity, creating a clunky rhythm. Similarly, in the Methodology section, the explanation of the Rectified Flow negation includes unnecessary connective words ("thus") that disrupt the causal flow. The description of the unified task representation in Section 3.2 is technically clear but overly verbose; it could be tightened significantly without losing precision.

A few typos and grammatical slips appear in the Experiments section. Specifically, "Qualatatively" is misspelled, and the phrase "are exceeding the quality" is non-idiomatic. Additionally, a sentence in the Emergent Behaviors subsection begins with a participle phrase ("trained on synthetic data...") but lacks the necessary conjunction or subject to form a complete thought, resulting in a sentence fragment.

These issues are minor and easily fixable with a careful copy-edit. They do not obscure the scientific contribution, but addressing them would improve the reading experience and ensure the prose matches the high quality of the research.

Action items.

- [**writing**] The paper is generally well-structured and the core argument flows logically from the problem statement to the proposed solution. However, several sentences suffer from redundancy, awkward phrasing, or minor grammatical errors that force the reader to pause and re-parse. In the Introduction, the authors repeat the concept of "unified" and "model" in close proximity, creating a clunky rhythm. Similarly, in the Methodology section, the explanation of the Rectified Flow negation includes unnecessar